

# scripts/run-challenge-matrix.sh — User Guide

**Last verified:** 2026-05-16 (HelixQA Option 1 wiring landed — --include-helixqa flag added) **Inheritance:** HelixConstitution §11.4.18 + Lava §6.AE.6 + §6.X (Container-Submodule Emulator Wiring) + §6.I (Multi-Emulator Container Matrix)

## Overview

Operator entry point for §6.AE gate-mode Challenge Test runs. Pre-bakes the §6.AE.2 minimum AVD matrix (API 28 / 30 / 34 / latest stable × phone + tablet) and delegates to submodules/containers/cmd/emulator-matrix --runner=containerized per §6.X.

## Honest pre-flight

The script DETECTS the §6.X-debt darwin/arm64 host gap (no /dev/kvm available to podman containers; macOS HVF not exposed to podman) and, when detected, REFUSES to claim a successful gate run. Instead it:

1. Validates arguments + matrix minimum
2. Writes <evidence-dir>/host-preflight.json with the host-gap classification
3. Exits 2 (NOT 0) — gate-host ineligible

Per §6.J/§6.L: no false-pass; honest unblock report. Real gate runs require a Linux x86\_64 + KVM host.

## Usage

```
# Run ALL Challenges on the §6.AE.2 minimum matrix
bash scripts/run-challenge-matrix.sh
```

```
# Run a specific Challenge
bash scripts/run-challenge-matrix.sh \
  --test-class
    lava.app.challenges.Challenge01AppLaunchAndTrackerSelectionTest
```

```

# Add TV-class AVD (when feature touches TvActivity)
bash scripts/run-challenge-matrix.sh --add-tv

# Add foldable AVD
bash scripts/run-challenge-matrix.sh --add-foldable

# Override "latest stable" API level
bash scripts/run-challenge-matrix.sh --latest-api 36

# Skip APK rebuild
bash scripts/run-challenge-matrix.sh --no-build

# Override the matrix entirely with explicit AVDs (target EXISTING
  host AVDs).
# Use on a macOS host-direct+HVF run where the default CZ_API*
  images are not provisioned.
bash scripts/run-challenge-matrix.sh --avds
  "CZ_API35_Phone_Fresh:35:phone" --no-build

# ALSO invoke the 11 HelixQA Challenge scripts (Option 1 wiring)
bash scripts/run-challenge-matrix.sh --include-helixqa

```

## Inputs

| Arg                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --test-class <fqcn>               | Specific Challenge to run (default: all under lava.app.challenges)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --evidence-dir <dir>              | Output directory (default: dated under .lava-ci-evidence/)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --no-build                        | Skip the :app:assembleDebug step                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| --avds<br>"<name:api:form[,...]>" | <b>Replace</b> the §6.AE.2 default matrix entirely with an explicit comma-separated AVD list. Use to target <b>existing host AVDs</b> (e.g. CZ_API35_Phone_Fresh:35:phone on a macOS host-direct+HVF run where the default CZ_API* images are not provisioned). When set, --latest-api / --add-tv / --add-foldable are ignored. A sub-minimum list is a <b>development-iteration</b> run, <b>not</b> a §6.AE.2-conformant gate matrix. NOTE: the Containers CLI still resolves each AVD's system image through tools/lava-containers/vm-images.json (--image-manifest); the manifest must contain a matching image id (e.g. android-35-phone) or the runner fails with no image with id. |

| Arg                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--latest-api &lt;N&gt;</code> | Override the "latest stable" API level (default: 36; ignored when <code>--avds</code> is set)                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>--add-tv</code>               | Add a TV-class AVD to the matrix                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>--add-foldable</code>         | Add a foldable AVD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>--include-helixqa</code>      | ALSO invoke <code>scripts/run-helixqa-challenges.sh</code> (the 11 HelixQA Challenge scripts per Option 1 wiring). OFF by default so existing matrix runs are unaffected. HelixQA runs on the HOST BEFORE the AVD matrix → independent of §6.X-debt darwin/arm64 gate-host gap. HelixQA wrapper evidence lands at <code>&lt;evidence-dir&gt;/helixqa/</code> . Non-zero HelixQA exit promotes the final aggregate exit code (matrix exit dominates if both fail). See <code>docs/scripts/run-helixqa-challenges.sh.md</code> for full details. |

## §6.AE.2 mandatory minimum matrix

```
CZ_API28_Phone:28:phone
CZ_API30_Phone:30:phone
CZ_API34_Phone:34:phone
CZ_API34_Tablet:34:tablet
CZ_API<latest>_Phone:<latest>:phone
```

Plus TV / foldable when explicitly requested by `--add-tv` / `--add-foldable`.

Sub-minimums are permitted for development iteration; the gate row's `gating: true` flag is only set when the full minimum is satisfied + every config dimension (theme/locale/density per §6.AE.2) is covered.

## Outputs

- `<evidence-dir>/host-preflight.json` — host-gap classification (always written)
- `<evidence-dir>/real-device-verification.{md,json}` — per-AVD attestation rows (only on Linux x86\_64 + KVM gate-host)
- `<evidence-dir>/helixqa/helixqa-attestation.json` + per-script logs — only when `--include-helixqa` is set

## Exit codes

| Code | Meaning                                                                             |
|------|-------------------------------------------------------------------------------------|
| 0    | Matrix completed; all AVDs passed                                                   |
| 1    | At least one AVD failed boot OR at least one test failed (real Containers CLI exit) |
| 2    | Gate-host ineligible (darwin/arm64 OR no KVM); honest unblock-needed report written |

## Cross-references

- `submodules/containers/cmd/emulator-matrix` (the underlying runner)
- `tools/lava-containers/vm-images.json` (matrix manifest)
- `scripts/run-emulator-tests.sh` (older sister-glue; same delegation, different default arguments)
- `scripts/run-helixqa-challenges.sh` + `docs/scripts/run-helixqa-challenges.sh.md` (the HelixQA wrapper invoked by `--include-helixqa`)
- `docs/plans/2026-05-16-helixqa-integration-design.md` (Option 1 wiring design)
- `.lava-ci-evidence/sixth-law-incidents/2026-05-13-emulator-container-darwin-arm64-gap.json` (the standing §6.X-debt incident)
- `Lava CLAUDE.md` §6.AE + §6.X + §6.I